A certified lower bound for the ovals problem

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Abstract

We derive a universal lower bound greater than 0.96221 for the lowest curvature Schrödinger eigenvalue of a smooth closed curve of length 2π. Projection averaging and a variation constraint yield an explicit dual certificate with outward-rounded arithmetic. The conjectured sharp bound one in AIM 1034 remains open.

Repository target. This manuscript concerns AIM problem 1034 at revision `3fa50884d129`. Its scope is stated below.

Status

This note does **not** establish the conjectured lower bound 1. It gives a complete derivation of the explicit lower bound

$$\lambda_1\left(-\frac{d^2}{ds^2}+\kappa(s)^2\right)\geq C_0>0.96221.$$

$$C_0\in(0.962217776056501568384828,0.962217776056501568384870).$$

The curve has length 2π; the bound holds for smooth closed curves in $\mathbb{R}^m$, $m\geq 2$, including the spatial curves in the problem. The numerical enclosure is certified by `verify_ovals_bound.py`, using only outward-rounded standard-library Decimal arithmetic and explicit series remainders. The proof relies on Denzler's established reduction of the global variational problem to smooth strictly convex planar minimizers, and on the projection-averaging inequality of Suragan. The latter is also derived below for clarity. No claim of priority or independent peer review is made.

The cited September 2026 bound is approximately 0.9618316523. The additional input here is a total-variation constraint on the antipodal arclength profile. The bound below remains strictly below 1, so problem 1034 stays unresolved.

1. Geometric reduction

Denzler proves that the infimum of the curvature Schrödinger eigenvalue among closed curves of length 2π is realized by a planar, analytic, strictly convex curve with positive curvature; his relaxed formulation also permits a digon of quotient 1. Consequently, a lower bound below 1 proved for smooth strictly convex planar curves holds for all the curves in the original problem.

This is Theorem 1 and Theorems 6, 8, 9 in the arXiv version of his paper, with renumbering in the published version.

We therefore consider an oriented strictly convex planar loop with tangent

$$T(s) = (\cos\phi(s), \sin\phi(s)), \qquad \phi'(s) = \kappa(s) > 0.$$

Let $s = \sigma(\theta)$ be the inverse tangent-angle coordinate and put $g(\theta) = \sigma'(\theta)$. Then

$$g > 0, \qquad \int_0^{2\pi} g(\theta)\, d\theta = 2\pi, \qquad \int_0^{2\pi} e^{i\theta} g(\theta)\, d\theta = 0. \tag{1}$$

The last identity is closure of the original loop. Define the antipodal arclength

$$\ell(\alpha) = \sigma(\alpha + \pi) - \sigma(\alpha) = \int_\alpha^{\alpha+\pi} g(\theta)\, d\theta. \tag{2}$$

It obeys three useful constraints:

$$\ell(\alpha) + \ell(\alpha + \pi) = 2\pi, \tag{3}$$

$$\int_0^\pi \ell(\beta - t) \sin t\, dt = 2\pi, \tag{4}$$

and, on every interval I of length π,

$$\mathrm{TV}_I(\ell) = \int_I |g(\alpha + \pi) - g(\alpha)|\, d\alpha \le \int_I (g(\alpha + \pi) + g(\alpha))\, d\alpha = 2\pi. \tag{5}$$

For completeness, (4) follows by Fubini from the complex identity

$$\int_0^\pi \ell(\beta - t) e^{it}\, dt = 2\pi i.$$

Indeed, the inner kernel after reversing the integrals is $i(1 + e^{i(\beta-\theta)})$; the second term integrates to zero by (1). The imaginary part is (4). Constraint (5) is the extra information used in this note.

2. Projection averaging

For a real $\psi \in H^1_{\mathrm{per}}(0, 2\pi)$, set $Y(s) = \psi(s)T(s)$. Then

$$|Y'(s)|^2 = |\psi'(s)|^2 + \kappa(s)^2 \psi(s)^2. \tag{6}$$

For each angle α, the scalar projection $Y(s) \cdot (-\sin\alpha, \cos\alpha)$ vanishes at the two points with tangent angles α and $\alpha + \pi$. Apply the sharp Dirichlet interval Poincaré inequality separately on the two complementary arcs, whose lengths are $\ell(\alpha)$ and $\ell(\alpha + \pi)$, and integrate in α. The identity

$$\int_0^{2\pi} |v \cdot (-\sin\alpha, \cos\alpha)|^2\, d\alpha = \pi |v|^2$$

gives

$$\int_0^{2\pi} (|\psi'|^2 + \kappa^2\psi^2)\, ds \geq 2\pi \int_0^{2\pi} J(\phi(s))\psi(s)^2\, ds, \tag{7}$$

where

$$J(\beta) = \int_0^{\pi} \frac{\sin^2 t}{\ell(\beta - t)^2}\, dt. \tag{8}$$

To verify the factor in (7), one of the two arc contributions, after reversing the order of integration, is $\pi^2 \int J(\phi(s))\psi(s)^2\, ds$. The other is identical. Their sum is divided by π, the factor on the averaged left side. For complex ψ, apply the real inequality to its real and imaginary parts and add.

3. Symmetrizing the arclength profile

Fix β, and write

$$h(t) = \frac{\ell(\beta - t)}{\pi}, \qquad 0 \leq t \leq \pi.$$

Equations (3)–(5) give

$$h > 0, \quad h(0) + h(\pi) = 2, \quad \int_0^{\pi} h(t) \sin t\, dt = 2, \quad \int_0^{\pi} |h'(t)|\, dt \leq 2. \tag{9}$$

Set $\widetilde{h}(t) = (h(t) + h(\pi - t))/2$. Convexity of $x \mapsto x^{-2}$ yields

$$\int_0^{\pi} \frac{\sin^2 t}{h(t)^2}\, dt \geq \int_0^{\pi} \frac{\sin^2 t}{\widetilde{h}(t)^2}\, dt. \tag{10}$$

The symmetrized function still has the same weighted integral and no larger total variation. Let $L = \pi/2$ and let v be its restriction to $[0, L]$. By symmetry,

$$v(0) = 1, \qquad \int_0^{L} v(t) \sin t\, dt = 1, \qquad \int_0^{L} |v'(t)|\, dt \leq 1. \tag{11}$$

The right side of (10) equals twice

$$F(v) = \int_0^{L} \frac{\sin^2 t}{v(t)^2}\, dt. \tag{12}$$

4. A rigorous dual certificate

All parameters in the following definition are exact:

$$a = \frac{5467}{10000}, \qquad b = 2a, \qquad c = \frac{54893}{50000}, \qquad \eta = \frac{2}{c^3}. \tag{13}$$

Notice $0 < a < b < c$. Define

$$t_a = \arcsin((a/c)^3), \qquad t_b = \arcsin((b/c)^3),$$

and the positive comparison function

$$v_*(t) = \begin{cases} a, & 0 \le t \le t_a, \\ c(\sin t)^{1/3}, & t_a \le t \le t_b, \\ b, & t_b \le t \le L. \end{cases} \tag{14}$$

This is a comparison function only; no assertion that it satisfies (11) is needed. Put

$$r(t) = \eta \sin t - \frac{2\sin^2 t}{v_*(t)^3}.$$

It is nonnegative on $[0, t_a]$, zero on $[t_a, t_b]$, and nonpositive on $[t_b, L]$. Define

$$A = \int_0^{t_a} r(t)\,dt, \qquad B = -\int_{t_b}^{L} r(t)\,dt, \qquad F_* = F(v_*). \tag{15}$$

Both A and B are nonnegative. Let $q(t) = -\int_t^L r(s)\,ds$. Thus $q' = r$, $q(L) = 0$, $q(0) = B - A$, and

$$\|q\|_\infty = \max\{B, A - B\}. \tag{16}$$

For any v satisfying (11), integration by parts gives

$$\begin{aligned} \int_0^L r(t)v(t)\,dt &= -q(0) - \int_0^L q(t)v'(t)\,dt \\ &\ge A - B - \max\{B, A - B\} \\ &= -\max\{0, 2B - A\}. \end{aligned} \tag{17}$$

The supporting-line inequality for $x \mapsto \sin^2 t/x^2$ is

$$\frac{\sin^2 t}{v(t)^2} \ge \frac{3\sin^2 t}{v_*(t)^2} - \frac{2\sin^2 t}{v_*(t)^3}v(t) = \frac{3\sin^2 t}{v_*(t)^2} + (r(t) - \eta \sin t)v(t).$$

Integrate and use (11) and (17). This proves

$$F(v) \ge D := 3F_* - \eta - \max\{0, 2B - A\}. \tag{18}$$

It follows from (8), (10), and (12) that $J(\beta) \ge 2D/\pi^2$ for every β. Substitution into (7) proves the desired universal bound with

$$\boxed{C_0 = \frac{4D}{\pi}.} \tag{19}$$

5. Explicit evaluation and numerical certification

Let

$$I_a = \frac{t_a}{2} - \frac{\sin(2t_a)}{4}, \qquad I_b = \frac{L - t_b}{2} + \frac{\sin(2t_b)}{4}, \qquad I_m = \int_{t_a}^{t_b} (\sin t)^{4/3}\,dt.$$

Then the constants in (15) are

$$F_* = \frac{I_a}{a^2} + \frac{I_m}{c^2} + \frac{I_b}{b^2},$$

$$A = \eta(1 - \cos t_a) - \frac{2I_a}{a^3}, \qquad B = \frac{2I_b}{b^3} - \eta \cos t_b. \tag{20}$$

The certificate encloses these values as follows (digits below are abbreviated, not the full outward intervals):

Quantity	Approximate value
A	0.00384988849391648837
B	0.00192548312815568169
D	0.75572407410315354459
C_0	0.96221777605650156838

To avoid dependence on an uncertified special-function routine, put $r_a = a/c$, $r_b = b/c$, and $d_j = \binom{2j}{j}/4^j \le 1$. The two needed expansions are

$$\arcsin(r^3) = \sum_{j\ge 0} \frac{d_j r^{6j+3}}{2j+1},$$

$$I_m = \sum_{j\ge 0} \frac{3d_j}{6j+7}(r_b^{6j+7} - r_a^{6j+7}). \tag{21}$$

After n terms, valid upper bounds for the positive tails are, respectively,

$$\frac{r^{6n+3}}{(2n+1)(1-r^6)}, \qquad \frac{3r_b^{6n+7}}{(6n+7)(1-r_b^6)}. \tag{22}$$

The program uses $n = 2000$ and 65-digit outward rounding. It encloses π by the alternating Machin formula and bounds the square roots $\cos t_a = \sqrt{1-r_a^6}$, $\cos t_b = \sqrt{1-r_b^6}$ using the documented correctly rounded Decimal square-root operation with a deliberately enlarged error interval. It obtains

$$0.96221777605650156838482818 14 < C_0 < 0.96221777605650156838486976 51. \tag{23}$$

In particular $C_0 > 0.96221$ is a rigorous numerical comparison.

6. Scope and the remaining gap

The projection-averaging estimate (7) still loses information about how close all scalar projections are to their first Dirichlet interval modes. The certificate improves only the pointwise lower bound on J. It therefore gives no justification for replacing C_0 by 1.

Numerically optimizing the analogous constrained scalar problem suggests a best constant around 0.9622183981, only slightly above the rational certificate used here. That numerical observation is not needed in the proof and is not asserted as a certified optimum. It suggests that a substantial further step must retain more than this pointwise profile information.

References

1. J. Denzler, *Existence and regularity for a curvature dependent variational problem*, Transactions of the AMS 367 (2015), 3829–3845. arXiv:1301.6322.
2. D. Suragan, *An Improved Bound for the Ovals Problem*, arXiv:2609.10775v1, September 2026. Proposition 3.1 supplies (7), Lemma 2.2 supplies (4), and Section 4 discusses the reduction to arbitrary spatial loops. Primary manuscript.